

花瓣仔的奇妙食物旅程

PETALINO'S AMAZING FOOD VOYAGE

第1頁 PAGE 1

準備好探索
營養的五個
過程嗎？

Ready to explore
the five
processes of
nutrition?

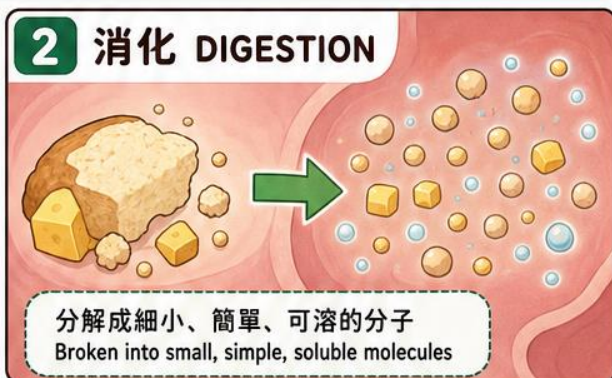


1 攝食 INGESTION



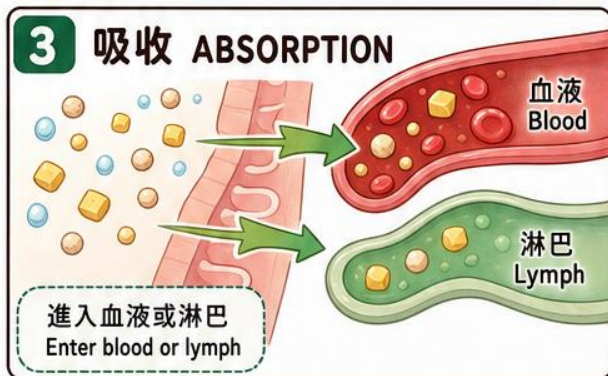
食物進入口腔
Food enters the mouth

2 消化 DIGESTION



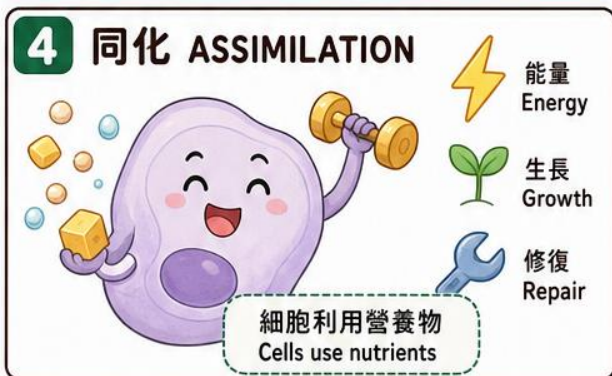
分解成細小、簡單、可溶的分子
Broken into small, simple, soluble molecules

3 吸收 ABSORPTION



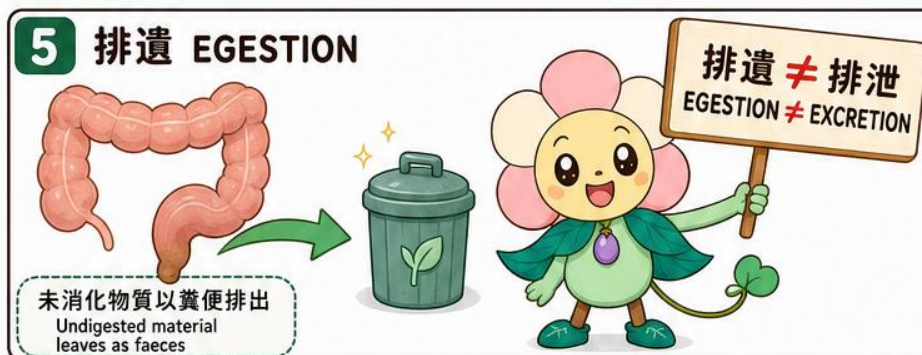
進入血液或淋巴
Enter blood or lymph

4 同化 ASSIMILATION



細胞利用營養物
Cells use nutrients

5 排遺 EGESTION



未消化物質以糞便排出
Undigested material
leaves as faeces

排遺 ≠ 排泄
EGESTION ≠ EXCRETION

! 排泄移除代謝廢物，
例如尿素和二氧化碳。
Excretion removes
metabolic waste,
e.g. urea and
carbon dioxide.



尿素
Urea



二氧化碳
Carbon dioxide

攝食 → 消化 → 吸收 → 同化 → 排遺
INGESTION → DIGESTION → ABSORPTION → ASSIMILATION → EGESTION

花瓣仔的奇妙食物旅程

PETALINO'S AMAZING FOOD VOYAGE

第2頁 PAGE 2
歡迎來到口腔
WELCOME TO THE MOUTH

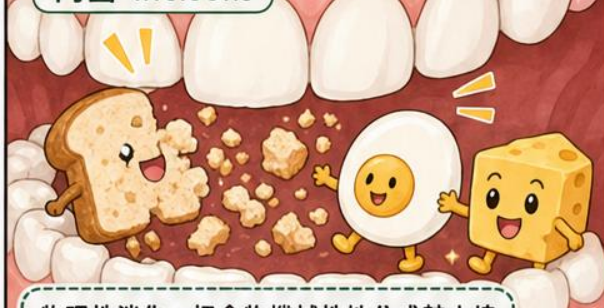
1 攝食 INGESTION



食物進入口腔
Food enters the mouth

2 咀嚼 MASTICATION

門齒 INCISORS



物理性消化：把食物機械性地分成較小塊
Physical digestion: mechanically breaks food into smaller pieces.

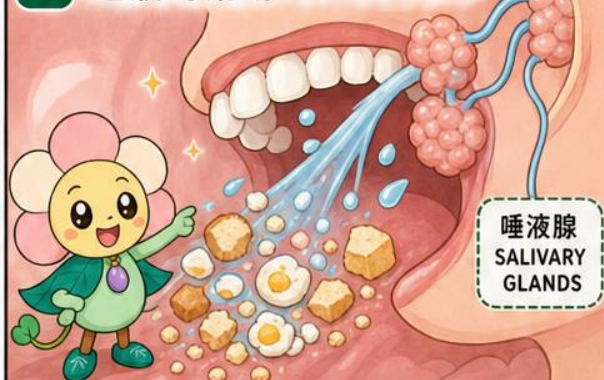
3 咀嚼 MASTICATION

臼齒 MOLARS



增加表面積，讓消化液更有效作用
Increases surface area for digestive juices.

4 唾液的幫助 SALIVA'S HELP



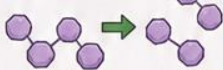
唾液腺
SALIVARY
GLANDS

5 唾液的神奇成分

MAGICAL SALIVA INGREDIENTS



唾液澱粉酶：澱粉 → 麥芽糖
Salivary amylase:
starch → maltose



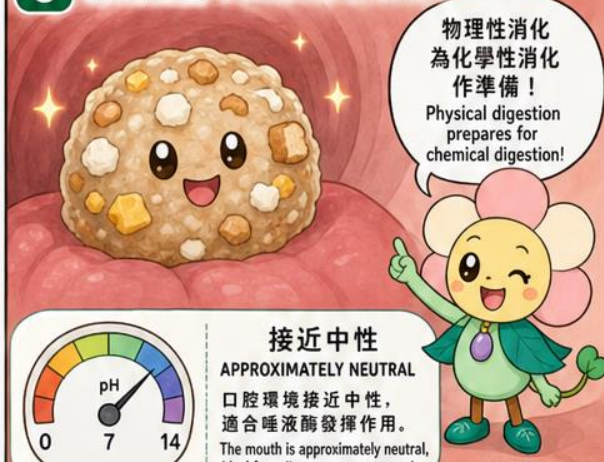
水：潤濕食物
Water:
moistens food



黏液：潤滑食物
Mucus:
lubricates food



6 形成食團 FORMING A BOLUS



物理性消化
為化學性消化
作準備！
Physical digestion
prepares for
chemical digestion!



接近中性
APPROXIMATELY NEUTRAL
口腔環境接近中性，
適合唾液酶發揮作用。
The mouth is approximately neutral,
ideal for salivary enzymes to work.

攝食
INGESTION



咀嚼
MASTICATION



唾液的幫助
SALIVA'S HELP



形成食團
FORMING A BOLUS



第3頁 PAGE 3 – 食道特快線 THE OESOPHAGUS EXPRESS

物理作用
PHYSICAL ACTION

蠕動：肌肉交替收縮和放鬆，
形成波浪式運動。

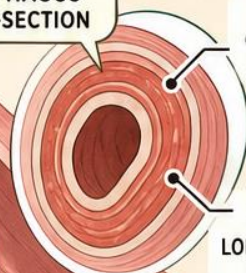
Peristalsis: muscles contract
and relax alternately,
creating a wave-like motion.

➡ 把食團朝一個方向
推向胃部。
Pushes the bolus towards
the stomach in one direction.



食團
BOLUS

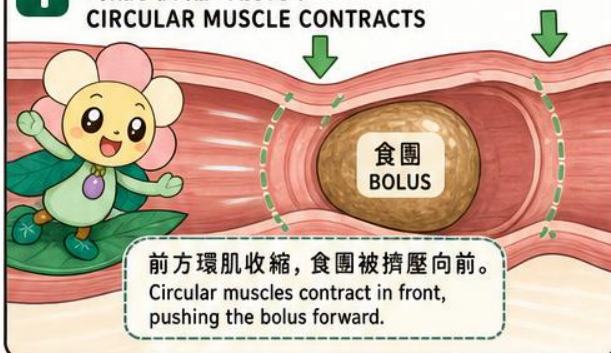
食道構造剖面圖
OESOPHAGUS
CROSS-SECTION



環肌
CIRCULAR
MUSCLE

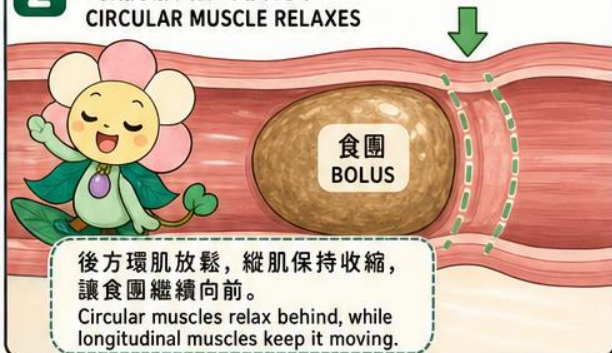
縱肌
LONGITUDINAL
MUSCLE

1 環肌收縮（前方） CIRCULAR MUSCLE CONTRACTS



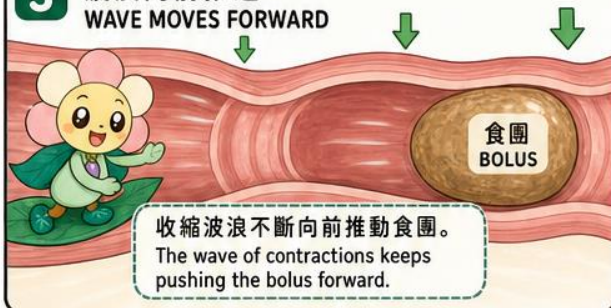
前方環肌收縮，食團被擠壓向前。
Circular muscles contract in front,
pushing the bolus forward.

2 環肌放鬆（後方） CIRCULAR MUSCLE RELAXES



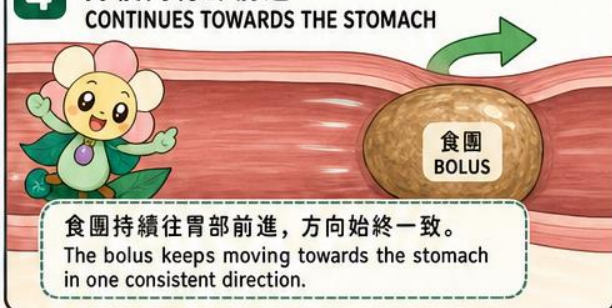
後方環肌放鬆，縱肌保持收縮，
讓食團繼續向前。
Circular muscles relax behind, while
longitudinal muscles keep it moving.

3 波浪向前推進 WAVE MOVES FORWARD



收縮波浪不斷向前推動食團。
The wave of contractions keeps
pushing the bolus forward.

4 持續向胃部前進 CONTINUES TOWARDS THE STOMACH



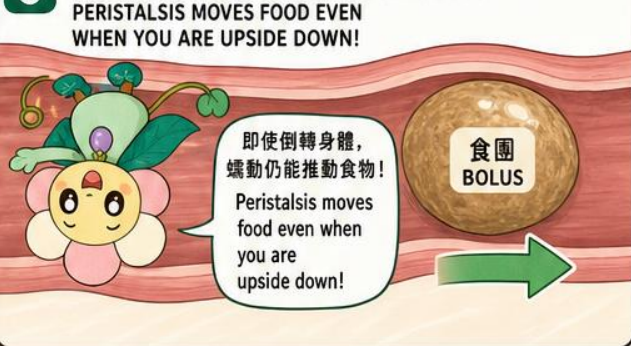
食團持續往胃部前進，方向始終一致。
The bolus keeps moving towards the stomach
in one consistent direction.

5 唾液澱粉酶繼續消化澱粉 SALIVARY AMYLASE CONTINUES DIGESTING STARCH



澱粉酶夥伴持續將澱粉分解成更小的糖分子。
Our amylase helper keeps breaking down
starch into smaller sugar molecules.

6 即使倒轉身體，蠕動仍能推動食物！ PERISTALSIS MOVES FOOD EVEN WHEN YOU ARE UPSIDE DOWN!



即使倒轉身體，
蠕動仍能推動食物！
Peristalsis moves
food even when
you are
upside down!



蠕動是食道的物理作用，不需消耗我們的能量，就能自動推動食物向前！

Peristalsis is a physical action of the oesophagus that doesn't use our energy—it works automatically to push food forward!

胃部酸性風暴

THE ACIDIC STOMACH STORM

1 花瓣仔抵達胃部！ PETALINO ARRIVES IN THE STOMACH!

蛋白質的
化學消化
開始了！
Chemical
digestion of
proteins has
began!



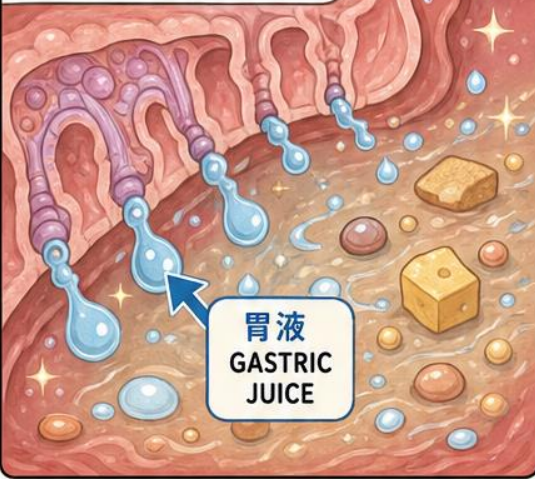
2 攪動 CHURNING

物理性地混合食物和胃液
Physically mixes food
with gastric juice.



3 胃腺分泌胃液 GASTRIC GLANDS RELEASE GASTRIC JUICE

胃液
GASTRIC
JUICE



4 胃部的三位守護好幫手 THREE AMAZING HELPERS IN THE STOMACH



胃蛋白酶
PEPSIN

蛋白質 → 多肽
proteins → peptides



鹽酸
HYDROCHLORIC
ACID

- 建立約 pH 2 的酸性環境
- 讓胃蛋白酶運作
- 殺死微生物
- 使蛋白質變性
- creates about pH 2
- allows pepsin to work
- kills microorganisms
- denatures proteins



黏液
MUCUS

保護胃壁免受胃酸和胃蛋白酶破壞
protects the stomach wall from acid and pepsin.

5 酸性環境：大約 pH 2 ACIDIC ENVIRONMENT: ~pH 2



酸性 ACIDIC 中性 NEUTRAL 鹼性 ALKALINE



6 食物的轉變 THE TRANSFORMATION

食團
BOLUS



食糜
CHYME



經過攪動與化學作用，食物變成食糜，準備進入小腸！

After churning and chemical action, food becomes chyme, ready to move into the small intestine!



攝食
INGESTION

消化
DIGESTION

吸收
ABSORPTION

同化
ASSIMILATION

排遺
EGESTION

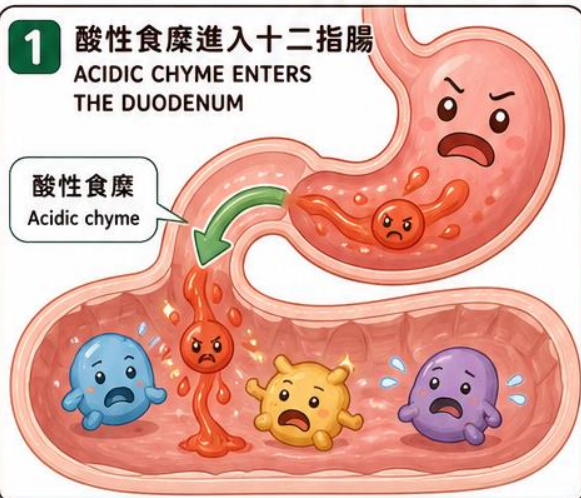
十二指腸中和任務

NEUTRALISATION IN THE DUODENUM

1 酸性食糜進入十二指腸

ACIDIC CHYME ENTERS THE DUODENUM

酸性食糜
Acidic chyme



2 花瓣仔啟動中和護盾

PETALINO ACTIVATES THE NEUTRALISING SHIELD

碳酸氫鈉
SODIUM
HYDROGENCARBONATE



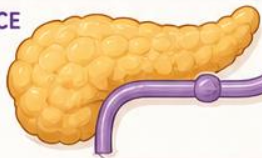
中和酸性食糜，建立較鹼性環境供酶運作
Neutralises acidic chyme and creates a relatively alkaline environment for enzymes.

3 三大支援軍團，齊聚十二指腸！

THREE SUPPORT SQUADS ARRIVE IN THE DUODENUM!

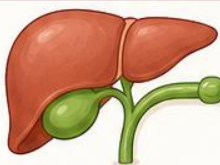
胰液 PANCREATIC JUICE

胰臟製造，
經胰管進入十二指腸
made by pancreas,
enters duodenum through
pancreatic duct



膽汁 BILE

肝製造，儲存在膽囊，
經膽管進入十二指腸
made by liver,
stored in gall bladder,
enters through bile duct



腸液 INTESTINAL JUICE

小腸壁細胞製造
produced by cells of
the small-intestine wall



腸液成分

COMPONENTS OF
INTESTINAL JUICE

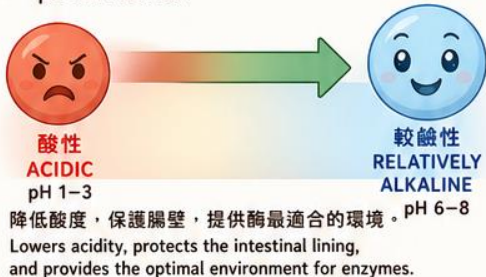
水
WATER

黏液
MUCUS

碳酸氫鈉
SODIUM
HYDROGENCARBONATE

4 酸性環境轉變為較鹼性環境

pH TRANSITION



5 中和完成，酶可以工作了！

NEUTRALISATION COMPLETE—ENZYMES CAN WORK!



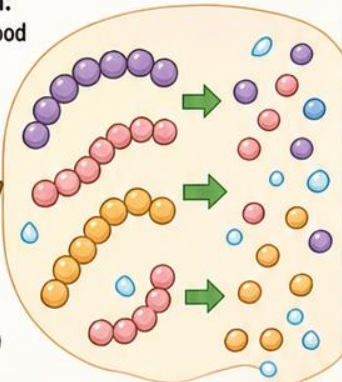
6 食物的路線圖

FOOD'S JOURNEY MAP



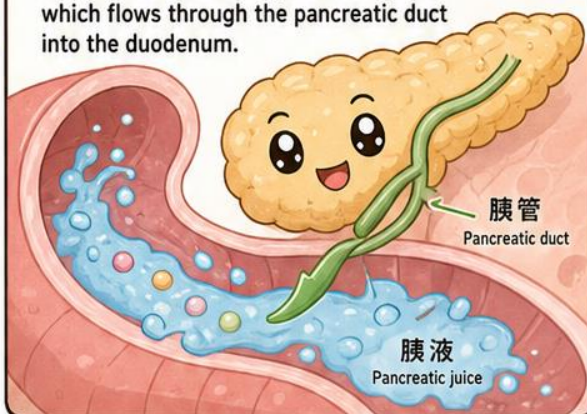
1 化學性消化：
酶把大食物分子分解成較簡單分子

Chemical digestion:
enzymes break large food
molecules into
simpler molecules.



2 胰臟分泌胰液，
經胰管進入十二指腸。

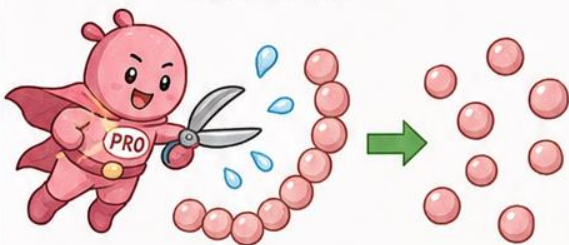
The pancreas secretes pancreatic juice,
which flows through the pancreatic duct
into the duodenum.



3 胰澱粉酶
PANCREATIC AMYLASE
消化碳水化合物
digests carbohydrates



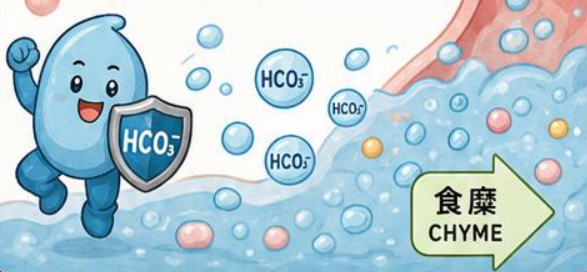
4 胰蛋白酶
PANCREATIC PROTEASE
消化蛋白質
digests proteins



5 胰脂肪酶
PANCREATIC LIPASE
消化脂質
digests lipids



6 碳酸氫鈉：維持較鹼性環境
Sodium hydrogencarbonate:
maintains a relatively alkaline environment.



各司其職，
開始化學性消化！
Different enzymes,
different jobs—chemical
digestion begins!



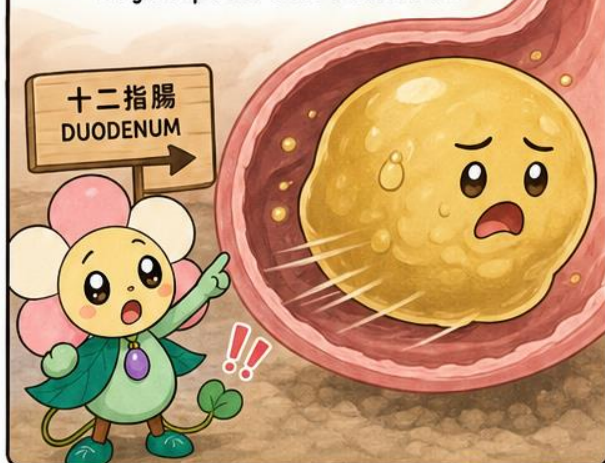
第7頁 — 膽汁對決巨型脂肪球

BILE VERSUS THE GIANT LIPID BLOB

第7頁 PAGE 7

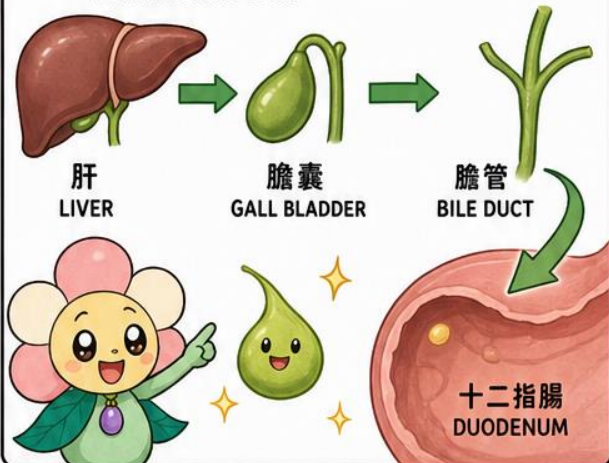
1 巨型脂肪球進入十二指腸！

The giant lipid blob enters the duodenum!



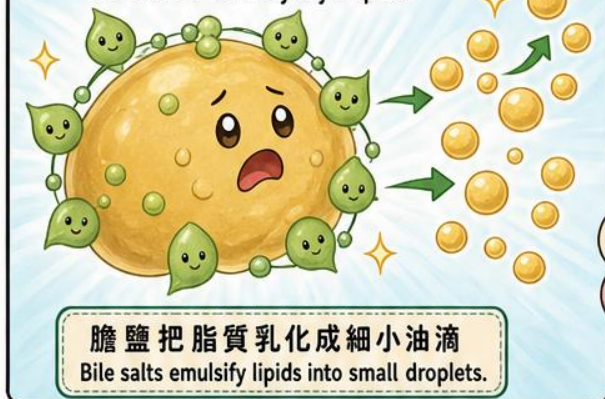
2 膽汁的正確路線

The correct route of bile



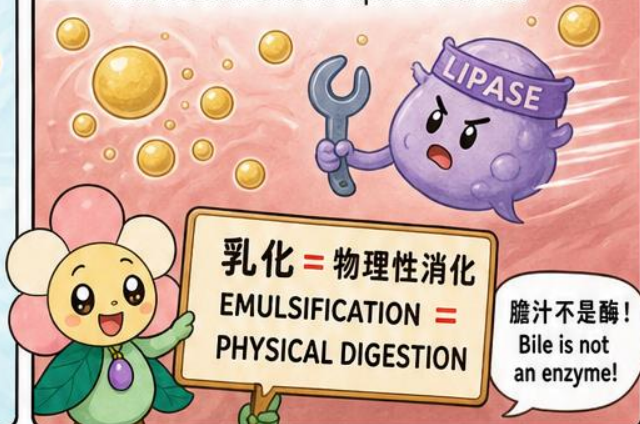
3 膽鹽包圍脂肪球，將它乳化成許多小油滴！

Bile salts surround the lipid blob and break it into many tiny droplets!



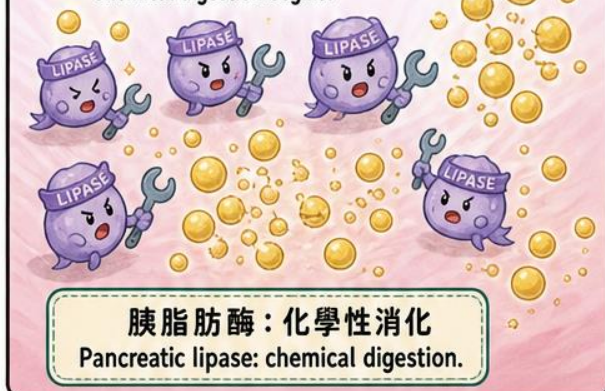
4 增加表面積，讓脂肪酶更快作用！

Increases surface area for lipase to act faster.



5 胰脂肪酶來了！開始化學性消化！

Pancreatic lipase attacks the droplets!
Chemical digestion begins!

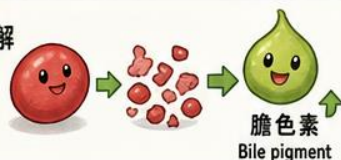


6 膽汁的兩大重要功臣！

Two more important roles of bile!

膽色素來自紅血球分解

Bile pigment comes from red blood cell breakdown.



碳酸氫鈉

幫助中和酸性食糜

Sodium hydrogencarbonate helps neutralise acidic chyme.



攝食
INGESTION

消化
DIGESTION

吸收
ABSORPTION

同化
ASSIMILATION

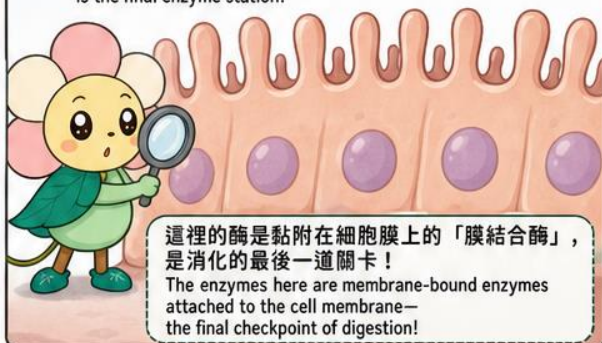
排遺
EGESTION



1 蠕動推動食糜通過迴腸。 Peristalsis pushes chyme through the ileum.

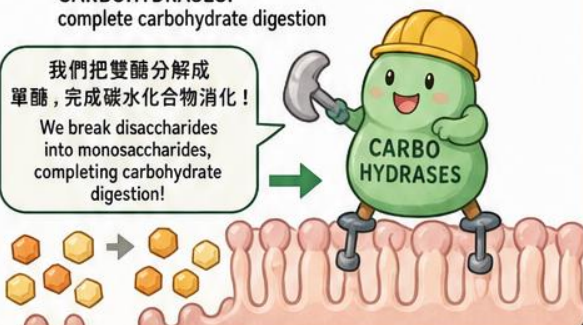


2 放大來看：小腸上皮細胞膜是最後的酶站！ Zoom in: the small-intestine epithelial cell membrane is the final enzyme station!



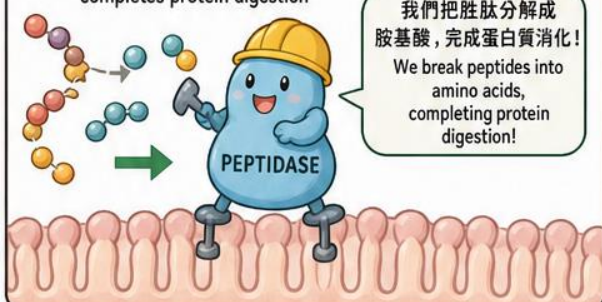
3 碳水化合物酶：完成碳水化合物消化 CARBOHYDRASES: complete carbohydrate digestion

我們把雙糖分解成單糖，完成碳水化合物消化！
We break disaccharides into monosaccharides, completing carbohydrate digestion!

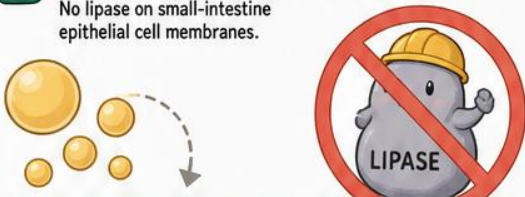


4 腸肽酶：完成蛋白質消化 PEPTIDASE: completes protein digestion

我們把胜肽分解成胺基酸，完成蛋白質消化！
We break peptides into amino acids, completing protein digestion!

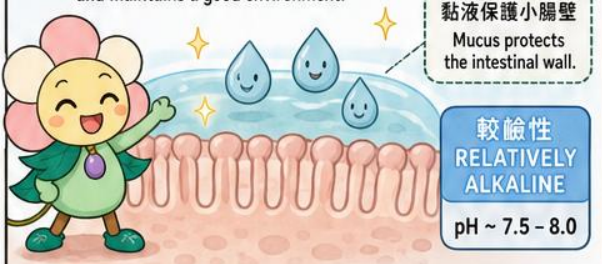


5 小腸上皮細胞膜上沒有脂肪酶。 No lipase on small-intestine epithelial cell membranes.



⚠ 脂肪的消化主要在小腸腔內由胰臟脂肪酶完成，而不是在細胞膜上。
Fat digestion is mainly completed in the intestinal lumen by pancreatic lipase, not on the cell membrane.

6 黏液保護小腸壁，維持良好環境。 Mucus protects the intestinal wall and maintains a good environment.



7 所有消化都在小腸完成！ All digestion is completed in the small intestine!



分清楚：胰臟酶 vs. 細胞膜酶

Make the distinction: Pancreatic enzymes vs. Cell membrane enzymes

胰臟酶（在腸腔內） Pancreatic enzymes (in the lumen)

負責大部分的消化工作
分解澱粉、蛋白質、脂肪
Do most of the digestion
Break down starch, protein, and fat

細胞膜酶（在細胞膜上） Cell membrane enzymes (on the cell membrane)

負責最終的分解步驟
完成碳水化合物與蛋白質的消化（無脂肪酶）
Handle the final breakdown
Complete carbohydrate and protein digestion (no lipase)

蠕動 推動食糜 → 胰臟酶 先分解 → 細胞膜酶 最終分解 → 養分 吸收利用

Peristalsis pushes chyme → Pancreatic enzymes start digestion → Cell membrane enzymes finish digestion → Nutrients absorbed and used

1 我們來到小腸絨毛！
準備進入吸收與同化之門，
把營養送到身體每個角落！
We've reached the villi
in the small intestine!
It's time to enter the gates
of absorption and assimilation
— delivering nutrients to
every part of the body!

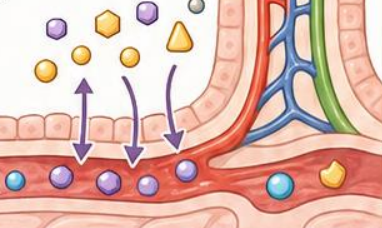


小腸絨毛結構 VILLI STRUCTURE

- 微血管
BLOOD CAPILLARY
- 乳糜管 (淋巴)
LACTEAL (LYMPH)
- 上皮細胞
EPITHELIAL CELLS

2 吸收 ABSORPTION

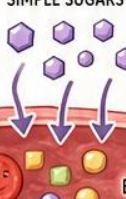
細小、簡單、可溶的分子穿過小腸壁，
進入運輸系統。
Small, simple, soluble molecules cross
the intestinal wall and enter the
transport system.



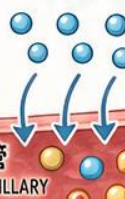
3 水溶性營養進入微血管 (血液)

Water-soluble nutrients enter
the blood capillary.

單糖
SIMPLE SUGARS



氨基酸
AMINO ACIDS



微血管
BLOOD CAPILLARY

4 脂質消化產物進入乳糜管 (淋巴)

Lipid digestion products enter
the lacteal (lymph).

脂肪酸 + 單甘油
FATTY ACIDS +
MONOGLYCERIDES



乳糜管 / 淋巴
LACTEAL / LYMPH



5 同化 ASSIMILATION

身體細胞攝取並利用營養物。
Body cells take up and use nutrients.



呼吸作用釋放能量
RESPIRATION
RELEASES ENERGY



生長
GROWTH

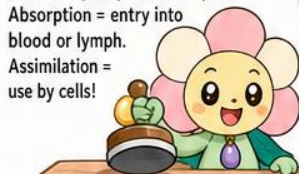


修補
REPAIR



6 營養護照蓋章！

吸收是進入血液或淋巴，
同化是細胞的使用！
Nutrient passports stamped!
Absorption = entry into
blood or lymph.
Assimilation =
use by cells!



單糖護照
SIMPLE SUGAR
PASSPORT



已吸收
ABSORBED

氨基酸護照
AMINO ACID
PASSPORT



已吸收
ABSORBED

脂質護照
LIPID
PASSPORT



已吸收
ABSORBED

前往身體細胞！
TO BODY CELLS!



吸收：進入運輸系統 (血液或淋巴)

ABSORPTION: ENTRY INTO TRANSPORT SYSTEM (BLOOD OR LYMPH)

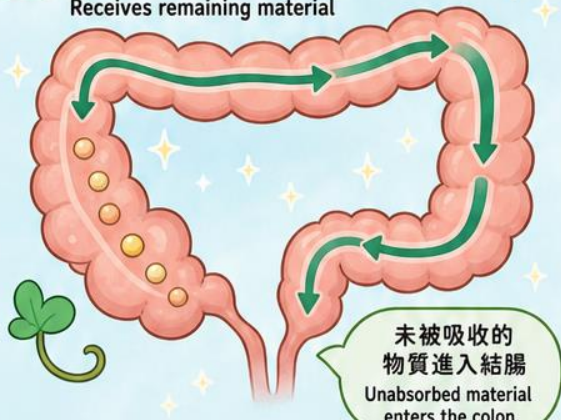


同化：細胞攝取並利用

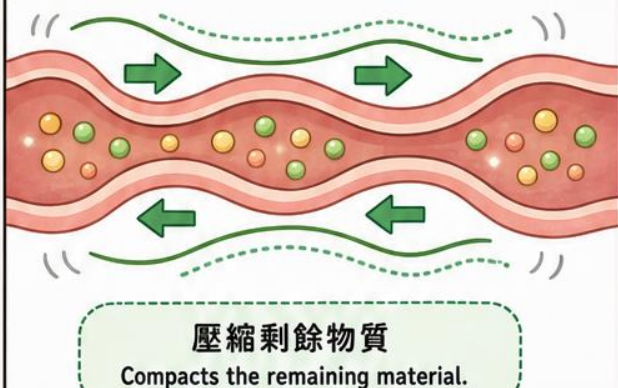
ASSIMILATION: CELLS TAKE UP AND USE NUTRIENTS



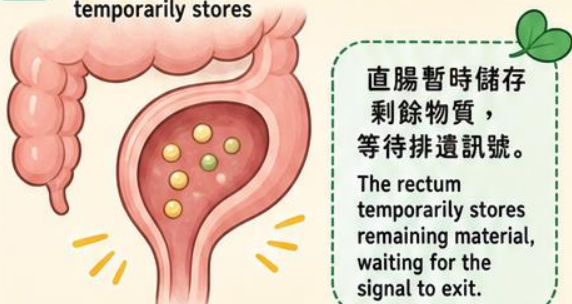
1 結腸 COLON：接收剩餘物質 Receives remaining material



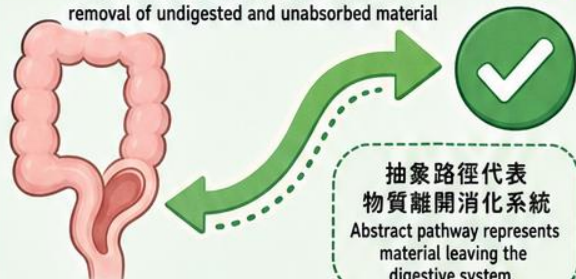
2 蠕動 PERISTALSIS



3 直腸 RECTUM：暫時儲存 temporarily stores



4 排遺 EGESTION： 移除未消化和未吸收的物質 removal of undigested and unabsorbed material



5 排遺不是排泄！ Egestion is not excretion!



6 小小科學筆記 SCIENCE NOTE

排泄移除代謝廢物。
Excretion removes metabolic waste.



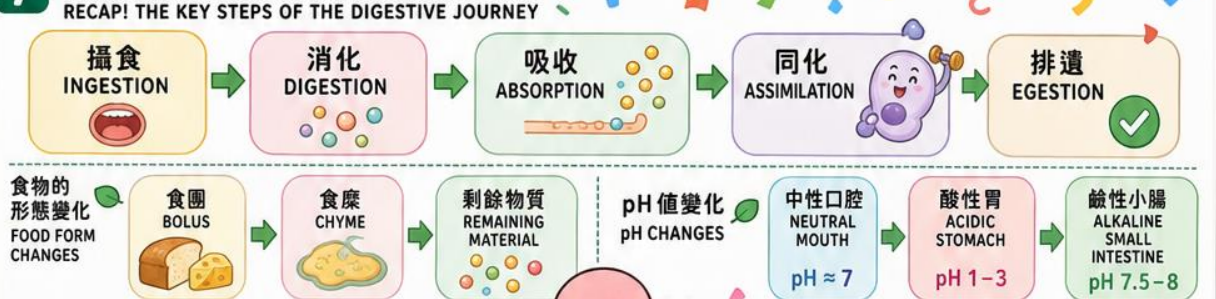
尿素
Urea



二氧化碳
Carbon dioxide



7 完整回顧！消化旅程的關鍵步驟 RECAP! THE KEY STEPS OF THE DIGESTIVE JOURNEY



8 任務完成！ Mission complete!

